

Comprehending and modifying ER diagrams and UML class diagrams

TASK: T2 TREATMENT: CD

First Name:.....

Last Name:.....

ID:.....

Start time:

Il seguente diagramma mostra il modello Entità/Relazioni di un sistema per la gestione di una azienda. In particolare il diagramma fa riferimento alla gestione delle risorse e dei progetti.

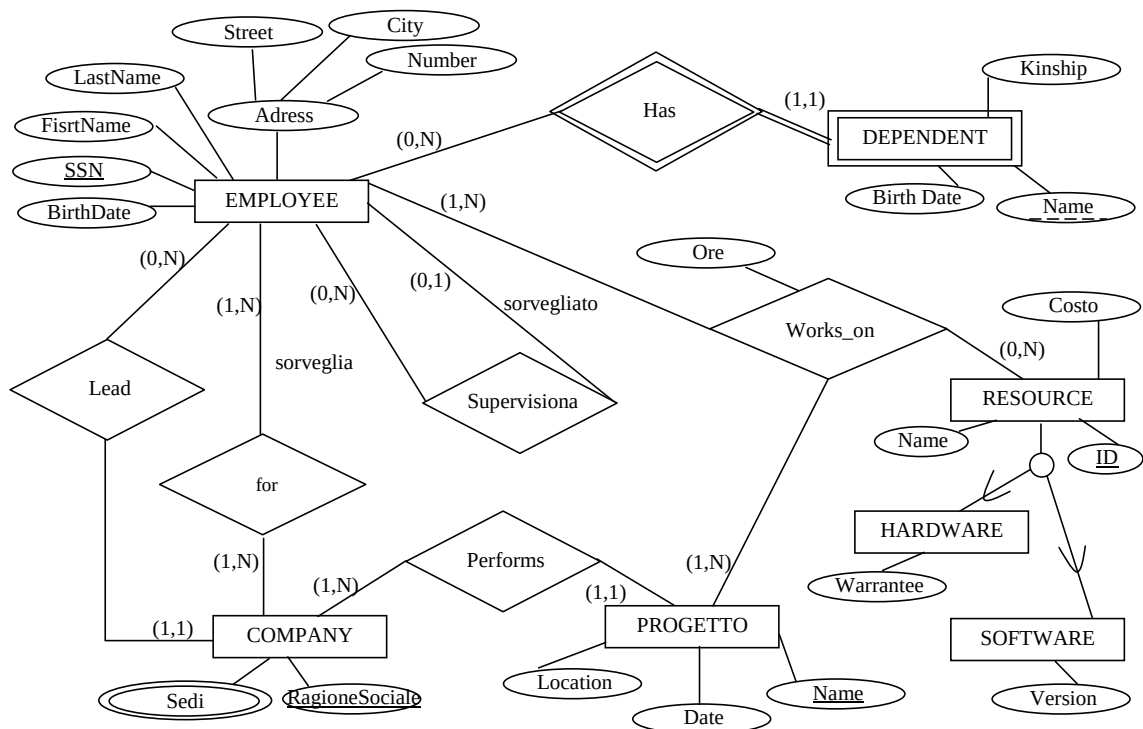
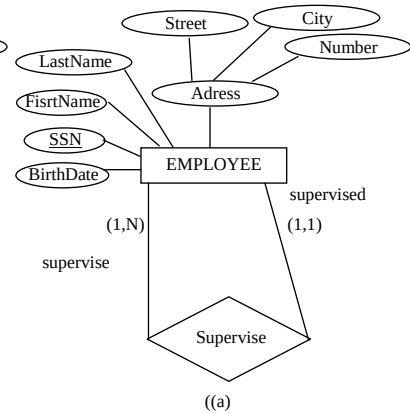
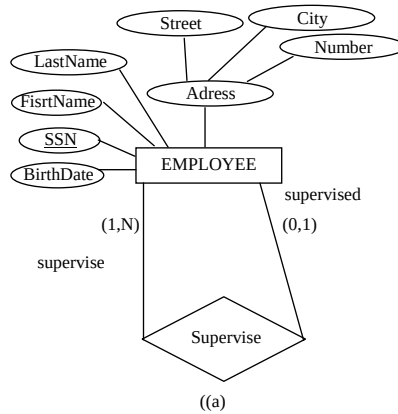
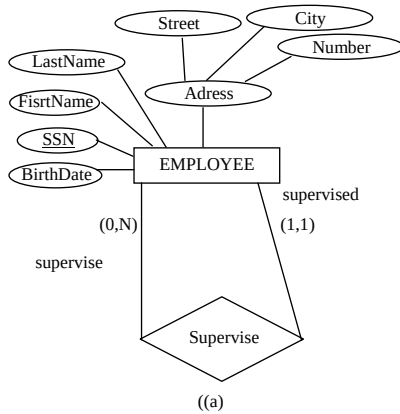


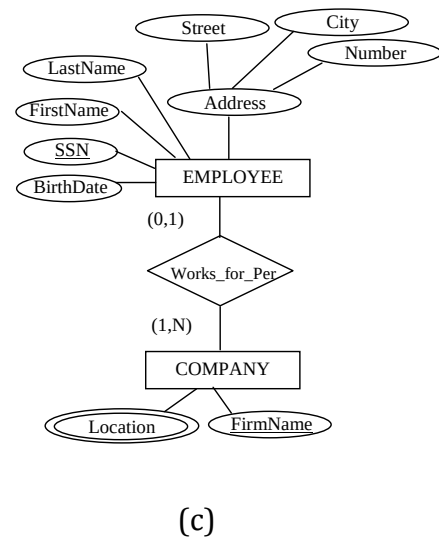
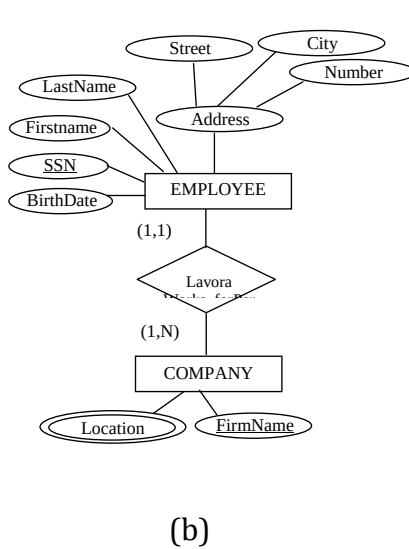
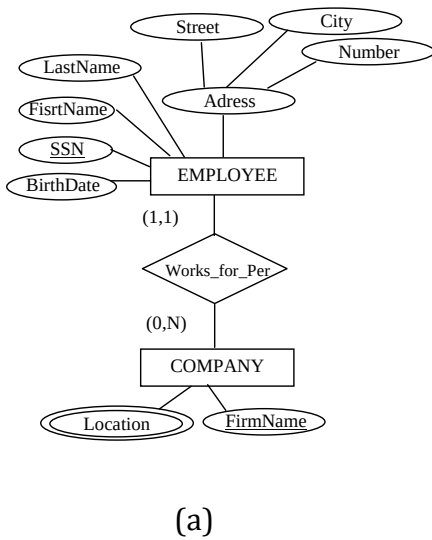
Figure 1: The ER diagram of a system managing company information

Analyze the diagram in Figure 1 and answer the following questions, regarding comprehension and modification activities.

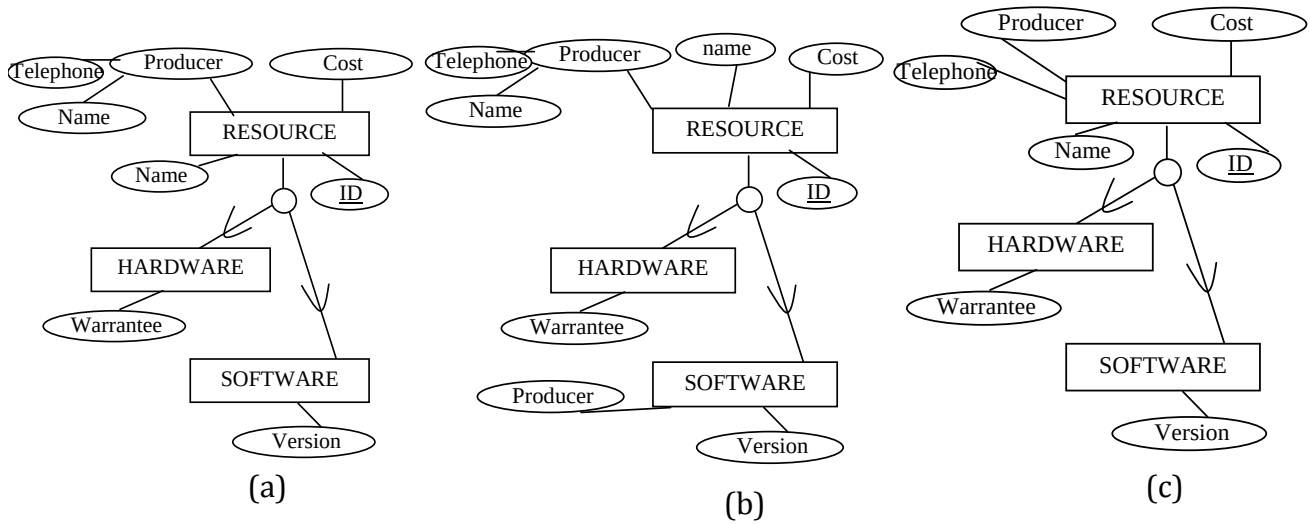
Q1 We want to specify a supervisor for each employee. An employee can be supervisor of several employees. Indicate the part of the ER diagram interested by this modification activity:



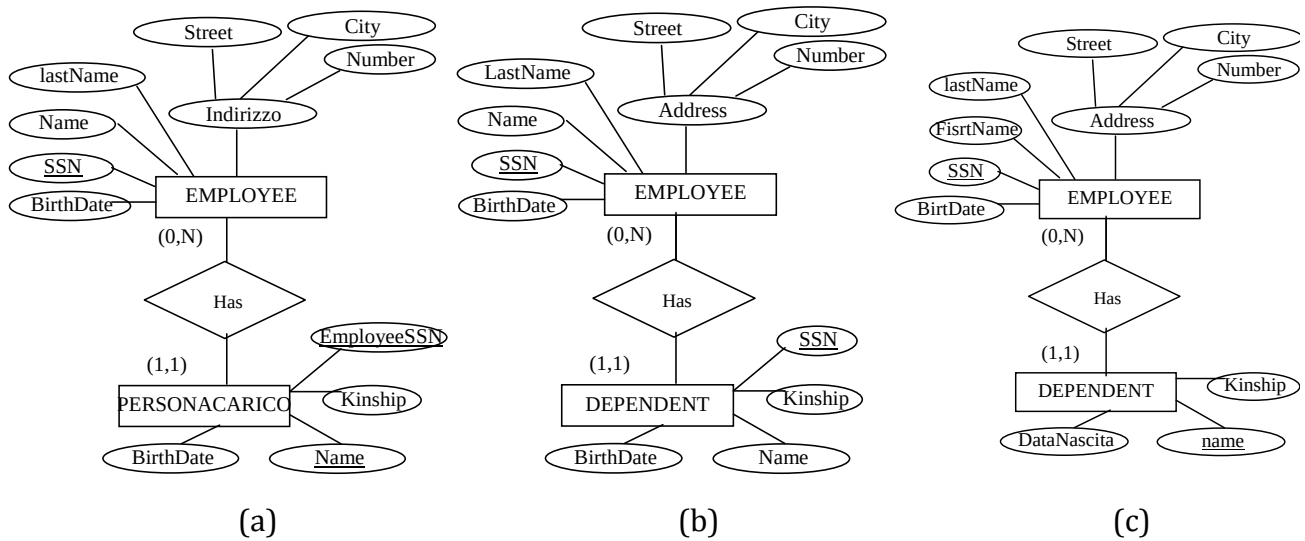
Q2 We want to specify that each employee works for at most one company. For each company can work one or more employees. Indicate the part of the ER diagram interested by this modification activity:



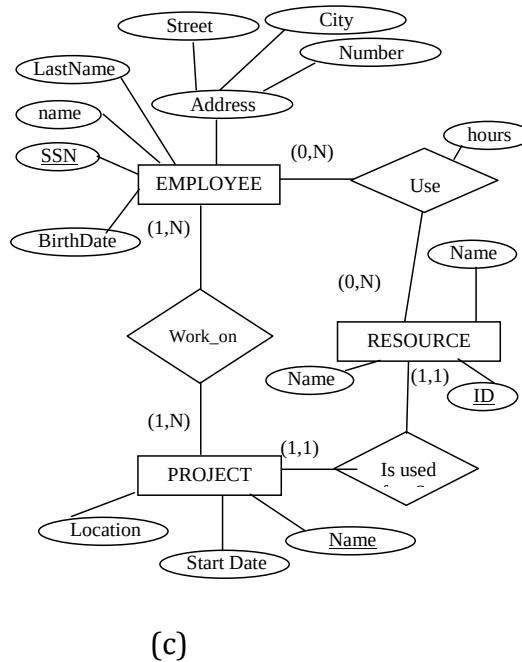
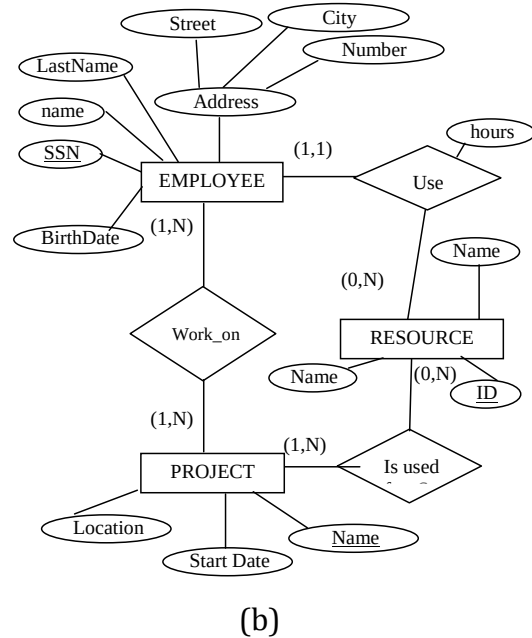
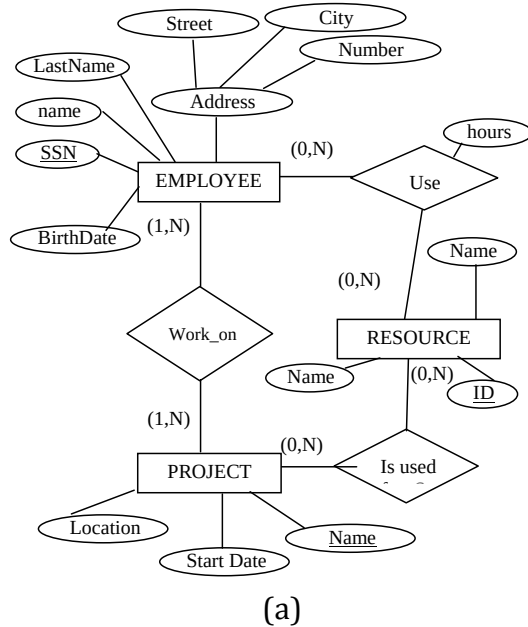
Q3 We want specify the information producer for each system. The information about the producer includes name and telephone. Indicate the part of the ER diagram interested by this modification activity:



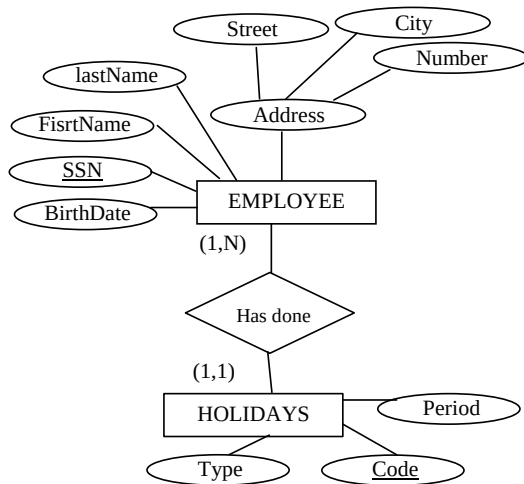
Q4 We want specify an identifier for the class dependent. Indicate the part of the ER diagram interested by this modification activity:



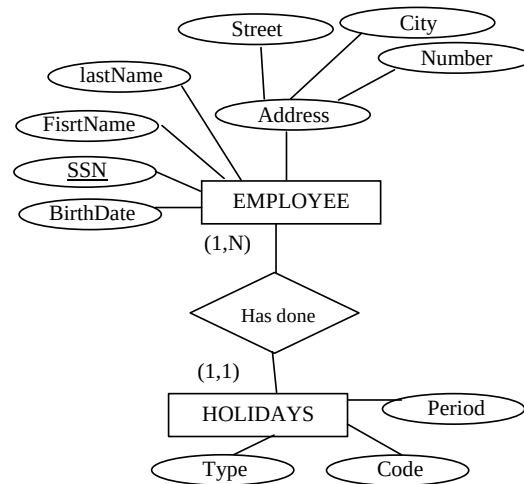
Q5 We want specify that an employee can use several resources for each project (he/she can also use no resources) and each resource can be used by several employees (a resource can be unused). A project can have several resources and each resource can be used in several projects. An employee works on at least one project and for each project at least one employee is involved. Indicate the part of the ER diagram interested by this modification activity:



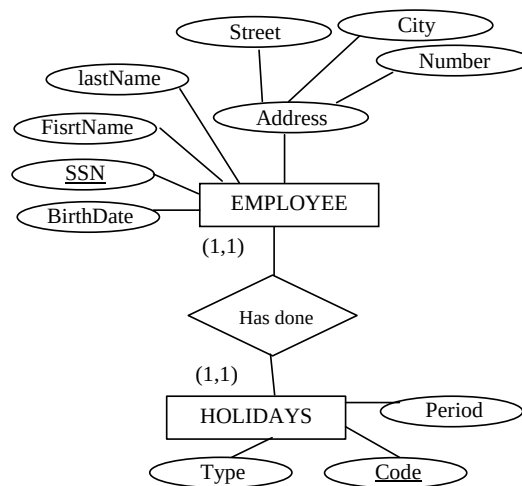
Q6 We want specify the information on holidays of each employee by providing period, type, and code (primary key). Indicate the part of the ER diagram interested by this modification activity:



(a)

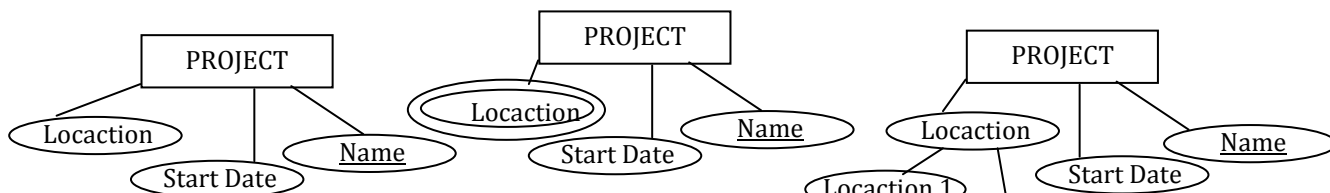


(b)

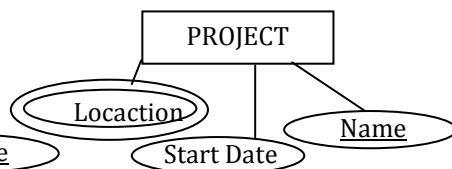


(c)

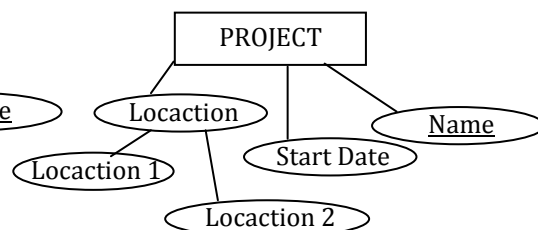
Q7 We want to specify that a project can involve several locations. Indicate the part of the ER diagram interested by this modification activity:



(a)

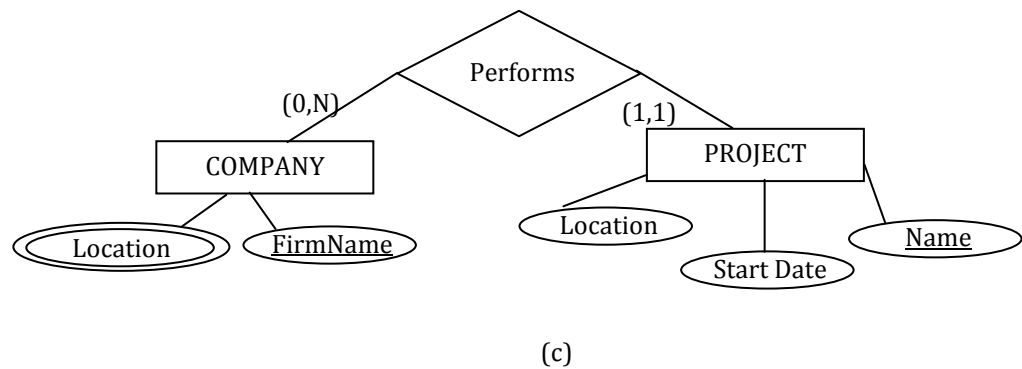
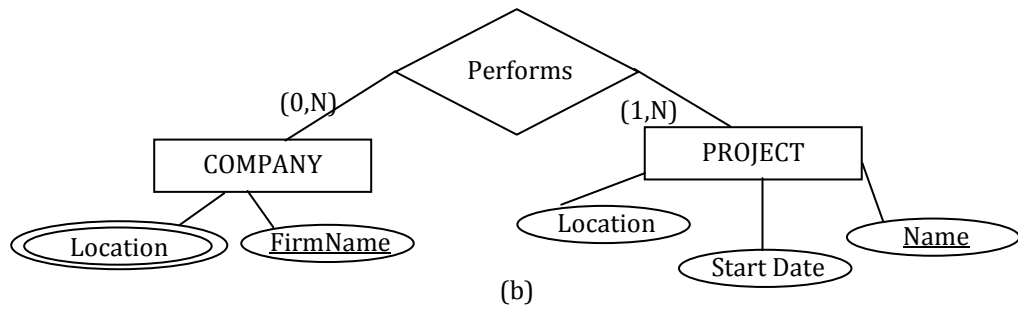
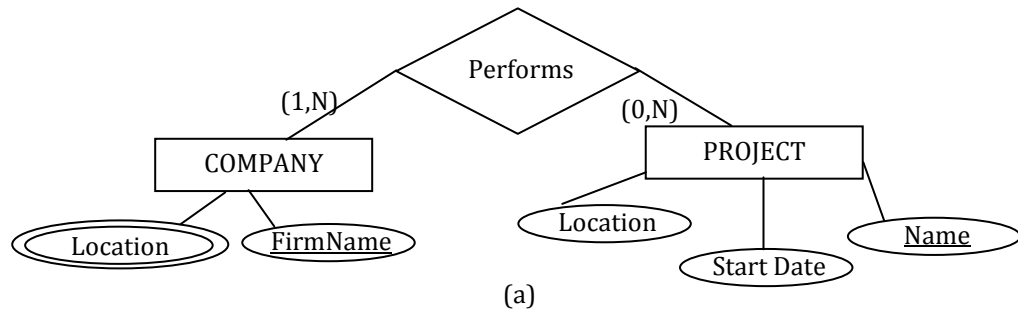


(b)

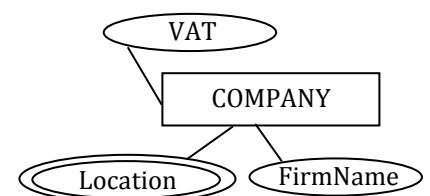
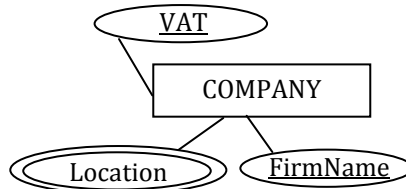
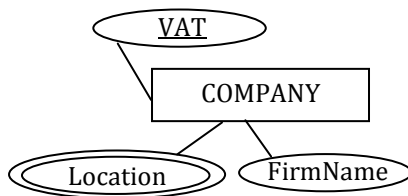


(c)

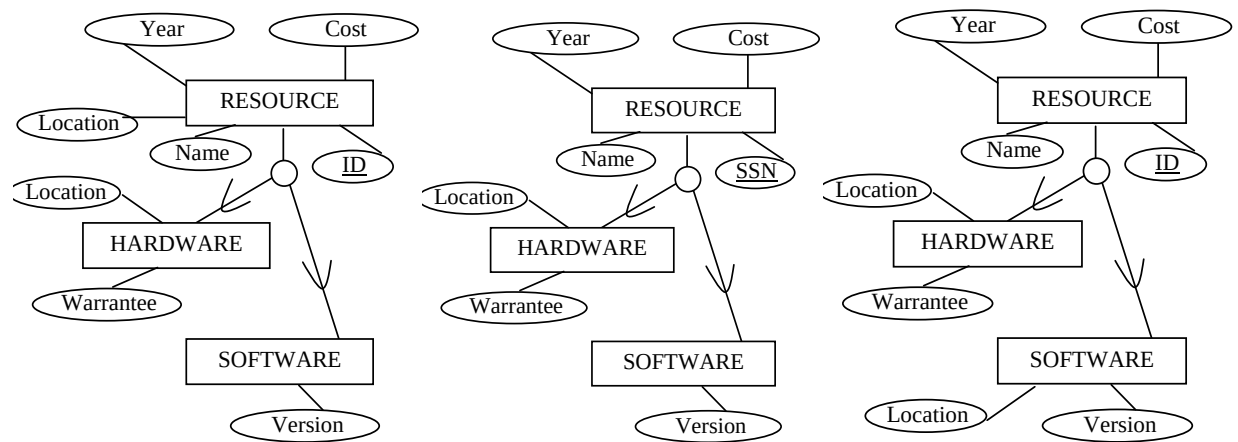
Q8 We want to specify that a company can be involved in no project.
Indicate the part of the ER diagram interested by this modification activity:



Q9 We want to identify a company by the information on the vat. Indicate the part of the ER diagram interested by this modification activity:



Q10 We want to specify the purchase date for each resource and the location of each hardware resource. Indicate the part of the class diagram interested by this modification activity:



End time:

Final questionnaire

The following questions have to answered after the execution of the comprehension and modification activities

A1 I had enough time to perform the tasks

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

A2. The task objectives were perfectly clear to me.

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

A3. The tasks I performed were perfectly clear to me.

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

A4. Judge the difficulty of the comprehension and modification task

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree